

LPI and Dry Lightning

Quick reference | BC continental HRDPS | 7 October 2026

WHAT THE LPI VALUE MEANS

LPI is a **0-100% model estimate of lightning occurring anywhere within 30 km** during the stated forecast period. LPI 40 means an estimated 40% chance of at least one lightning occurrence nearby. It does not describe flash density or the exact strike location. **24-hour and 3-hour probabilities are calculated separately.**

HOW THE MODEL COMBINES THE INGREDIENTS

Hourly ingredients become **20 summaries**: peaks, means, fractions of favourable hours, and combinations that occur together. Each summary is smoothed with a **20 km Gaussian sigma** on a 5 km analysis grid. The model combines these with **32 nonlinear ingredient patterns** and converts the learned weighted result to a probability.

WHAT GOES INTO THE LPI

Instability

Most-unstable lifted index (MU-LI): scaled from +1 to -5 C.
Most-unstable CAPE: 75-800 J/kg.

Charging environment

Humidity in the 0 to -20 C layer: 45-80%, weighted most strongly near -15 C. Its weighted pressure depth: 35-150 hPa. Mid-level humidity (-5 to -30 C): 35-75%.

These describe the potential for an unstable, humid mixed-phase cloud.

Storm development and rain

Resolved ascent at 500/700 hPa: 0.005-0.050 m/s. Trailing 3-hour rain: 0.05-1.5 mm. Rain rate: 0.02-0.8 mm/h.

Humidity and persistence

Surface and subcloud humidity; average instability, charging and ascent; fractions of hours with instability/ascent; same-hour storm ingredient combinations.

Ranges scale inputs to 0-1. Learned weights and patterns set their combined influence.

DRY-LIGHTNING SCREENING

Hourly dryness: $RH_{dry} = \min(\text{surface RH}, \text{subcloud RH})$. The dryness factor starts below **55% RH** and reaches full strength at **30% RH**. Subcloud RH averages surface RH and valid above-ground RH at 850/800/750/700 hPa.

Three-hour dry score: where 3-hour LPI is at least **20%**, multiply LPI by the strongest hourly dryness factor in that block. Reduce the score as trailing 3-hour rain rises from **0.25 to 2.5 mm**; the rain factor reaches zero at 2.5 mm.

Asterisks appear at a dry score of 15 or more. In the BC two-panel display they are grey below LPI 60% and black at 60% or more. This score is a **heuristic screen**, not a dry-lightning or fire-ignition probability. Check local rainfall, fuels and observations.

READING THE DISPLAY AND USING THE GUIDANCE

The right panel has fire-danger shading and purple-to-magenta LPI contours at **20, 40, 60 and 80%**. Cyan/teal dots show 3-hour rain at **2.5/10 mm**. The left panel shows humidity and gusts. The daily product covers **12Z to the next 12Z**; a three-hour map covers the block ending at its valid time. Confirm the period before interpreting a value.

Deployment scope: 12Z continental HRDPS, first 24 forecast hours. Experimental partial-season calibration. Missing ingredients remain missing; blank areas are not a forecast of zero lightning.